

Sky-Guard Radar Detection System

360° low-altitude UAV detection and precision EO cueing



Overview

A Ku-band radar subsystem composed of the radar unit, motorised turntable, and operator terminal software. It detects and tracks low-altitude UAV targets, outputs real-time 3D target parameters, and provides precise cueing for electro-optical, jamming, and spoofing subsystems.

Key Capabilities

- 360° mechanical azimuth scan with electronic elevation coverage from 0° to 30°.
- Outputs range, azimuth, altitude, speed, longitude, and latitude in real time.
- Supports tripod deployment, vehicle-roof mobile monitoring, or fixed rooftop stations.
- Fully coherent pulse-Doppler processing, monopulse angle measurement, MTD integration, and CFAR detection.
- Track initiation, target association, terminal reporting, and guidance output for EO identification.

Technical Specifications

Frequency Band	Ku, 16 GHz ± 100 MHz
Azimuth / Elevation	360° mechanical / ≥ 30° electronic
Max Detection Range	≥ 3 km (DJI Phantom 3/4, Pd = 85%)
Minimum Detection Range	≤ 250 m
Minimum Detectable Speed	≤ 1.5 m/s
3D Accuracy	Range ≤ 10 m; azimuth/elevation ≤ 0.5° RMS
Refresh Rate	3 s per 360° scan
Weight / Power	≤ 30 kg; ≤ 150 W including turntable

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The information contained in this document is provided for reference purposes only. Specifications are subject to change without prior notice.
Actual performance may vary depending on operating environment, deployment configuration, and integration conditions.